

FORT LIARD, NT, Canada

WMO: 714970

Lat: 60.236N

Lon: 123.473W

Elev: 213

StdP: 98.79

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.5	(c) -34.2	(d) -42.1	(e) 0.1	(f) -37.3	(g) -38.7	(h) 0.1	(i) -34.2	(j) 9.4	(k) -14.0	(l) 8.5	(m) -15.2	(n) 2.7	(o) 220	(p) 0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.9	(c) 28.6	(d) 17.2	(e) 26.6	(f) 16.4	(g) 24.9	(h) 15.6	(i) 19.1	(j) 25.5	(k) 17.9	(l) 24.1	(m) 16.9	(n) 22.7	(o) 2.3	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.2	21.4	15.6	11.4	20.2	14.5	10.6	19.1	55.1	25.5	51.5	24.1	48.3	22.9	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 7.3	(c) 6.4	(e) -40.2	(f) 32.0	(g) 3.5	(h) 1.7	(i) -42.8	(j) 33.2	(k) -44.8	(l) 34.2	(m) -46.8	(n) 35.2	(o) -49.3	(p) 36.4		
(a) 8.2	(b) 7.3	(c) 6.4	(e) -40.1	(f) 21.8	(g) 3.5	(h) 1.6	(i) -42.6	(j) 22.9	(k) -44.6	(l) 23.9	(m) -46.6	(n) 24.8	(o) -49.1	(p) 25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-21.2	-15.7	-9.9	2.0	10.3	15.3	17.5	15.6	9.7	0.8	-12.6	-19.1		
	DBStd	15.12	8.64	7.52	8.11	5.57	4.80	3.30	2.94	3.49	4.15	4.87	7.11	7.82		
	HDD10.0	4529	967	721	617	242	54	3	1	3	55	286	679	903		
	HDD18.3	6933	1225	954	875	491	251	99	49	97	259	542	929	1161		
	CDD10.0	686	0	0	0	1	63	163	234	177	46	2	0	0		
	CDD18.3	48	0	0	0	0	2	9	25	12	1	0	0	0		
	CDH23.3	743	0	0	0	0	69	188	307	164	14	1	0	0		
	CDH26.7	165	0	0	0	0	12	41	74	35	2	0	0	0		
(14) Wind	WSAvg	2.8	3.0	3.0	3.3	3.1	2.8	2.4	2.2	2.3	2.4	2.6	2.8	3.0		
(15) Precipitation	PrecAvg	443	26	20	16	18	41	68	78	54	50	30	32	25		
	PrecMax	622	79	76	44	54	84	141	240	108	107	82	89	72		
	PrecMin	345	5	6	1	1	6	20	9	20	8	7	3	7		
	PrecStd	77	22	15	9	13	21	37	50	26	30	20	20	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	8.1	13.0	19.6	28.3	30.3	31.7	30.2	24.9	17.5	8.1	10.1		
		MCWB	2.5	2.3	5.2	9.2	14.2	16.6	19.0	19.5	15.3	9.7	2.7	4.8		
	2%	DB	0.9	3.1	9.0	16.6	25.2	27.7	28.7	27.1	21.6	13.5	3.9	1.4		
		MCWB	-1.6	-0.8	3.2	7.6	13.0	16.0	17.9	17.8	12.9	6.9	0.4	-1.2		
	5%	DB	-5.7	-1.9	6.2	14.1	22.7	25.4	26.7	25.0	19.6	10.7	-0.5	-4.1		
		MCWB	-6.8	-3.8	1.4	6.0	12.0	15.0	17.2	16.5	12.0	5.7	-2.1	-5.2		
	10%	DB	-10.5	-5.5	2.3	11.9	20.1	23.4	24.7	23.0	17.2	7.8	-3.5	-8.7		
		MCWB	-11.2	-6.7	-0.8	4.9	10.6	14.2	16.4	15.6	10.9	4.4	-4.4	-9.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	2.5	5.8	9.7	15.6	18.9	20.8	20.9	16.2	10.4	2.7	5.2		
		MCDB	6.4	7.7	12.7	18.5	24.5	26.3	28.0	27.9	22.7	16.0	7.5	9.7		
	2%	WB	-1.2	-0.6	3.4	7.9	14.0	17.5	19.3	19.0	14.2	7.7	0.2	-0.8		
		MCDB	1.3	2.9	8.7	15.9	23.1	24.3	26.0	24.6	20.2	12.1	3.5	1.5		
	5%	WB	-6.6	-3.7	1.5	6.4	12.7	16.5	18.3	17.6	12.7	6.1	-2.3	-5.5		
		MCDB	-5.6	-1.8	5.9	13.2	21.1	22.9	24.5	23.3	17.7	10.4	-0.7	-4.2		
	10%	WB	-11.1	-6.6	-0.7	5.1	11.4	15.4	17.3	16.3	11.6	4.5	-4.6	-9.6		
		MCDB	-10.5	-5.4	2.5	11.4	19.1	21.4	23.1	21.5	16.1	7.6	-3.6	-9.0		
(35) Mean Daily Temperature Range	MDBR		7.4	10.4	12.9	12.7	13.6	12.6	11.9	11.9	11.1	7.5	7.0	7.2		
	5% DB	MCDBR	13.6	15.5	17.5	17.3	18.9	17.1	16.0	15.9	15.3	13.4	11.0	13.5		
		MCWBR	11.4	12.2	11.9	9.5	8.5	7.3	7.1	7.6	8.0	7.8	8.4	11.4		
	5% WB	MCDBR	13.7	15.2	17.1	16.5	17.3	14.6	14.5	14.7	14.3	12.5	10.6	13.5		
		MCWBR	11.3	12.1	11.6	9.3	8.1	6.8	7.1	7.7	8.0	7.5	8.0	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.212	0.245	0.272	0.335	0.349	0.363	0.370	0.361	0.315	0.287	0.223	0.177		
	taud		2.171	2.306	2.272	2.228	2.416	2.430	2.422	2.462	2.581	2.513	2.201	2.042		
	Ebn at Noon		598	787	868	860	878	869	851	831	825	733	567	484		
	Edh at Noon		38	62	89	114	103	103	102	91	68	52	36	27		
(44) All-Sky Solar Radiation	RadAvg		0.42	1.31	3.00	4.72	5.52	5.70	5.43	4.39	2.82	1.32	0.49	0.23		
	RadStd		0.03	0.10	0.20	0.21	0.37	0.43	0.43	0.35	0.29	0.12	0.06	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	+2.39	+3.44	N/A	N/A	N/A	-169	-231	N/A	+15
(47) Regional (5 neighbors)		N/A	+1.91	+2.50	N/A	N/A	N/A	N/A	N/A	+65	N/A

Nomenclature: See separate page